

Research Questions

Christoph Hanck

Summer 2024





What is a research question anyway?

Key Konzept: Research question

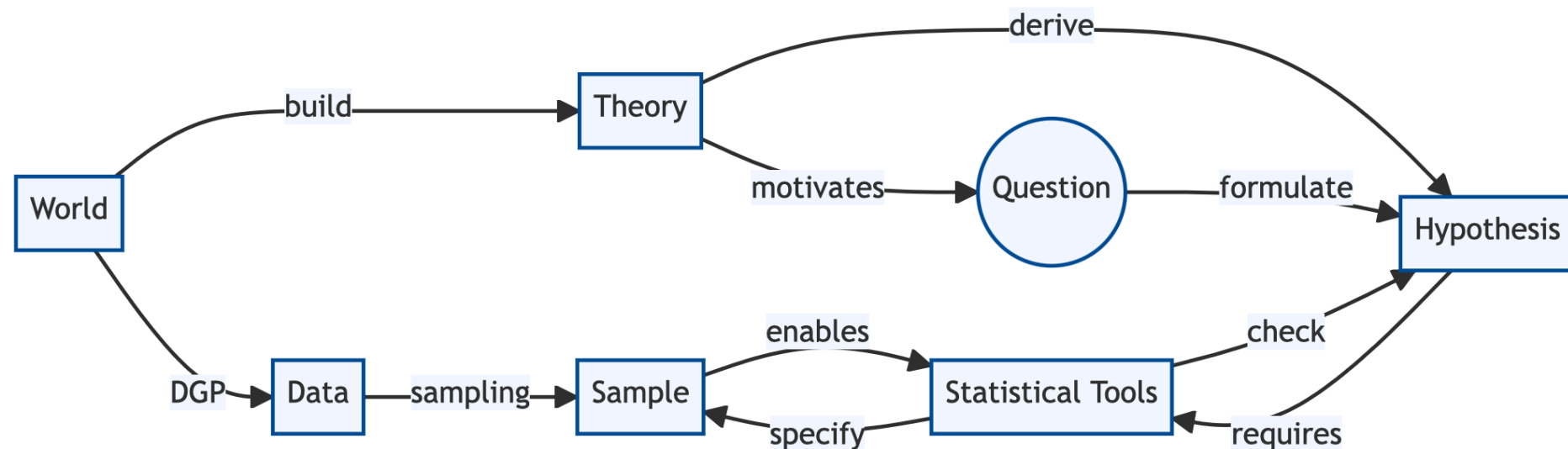
A **research question** is a question

1. ... that *can be answered* and
2. ... for which having that answer will *improve our understanding of how the world works*.

A research question takes us from a **theory** to an **hypothesis** that we can check using real world data. The data may provide **evidence** in favor or against our hypothesis.



What is a research question anyway? – ctd.





Research questions

Example: The influence of smartphone usage on children

- Does heavy smartphone usage damage children's eyesight?
- Does smartphone usage stimulate creative or intellectual activity?
- Does smartphone usage reduce children's ability to concentrate?



Why not simply look for patterns in the data?

- We are living in an information economy, surrounding us by an increasing amount of readily available data
- Why not skip hassling with theories and research questions and simply look for *patterns* in the data?
- A lot of people do — this commonly referred to as **data mining**

Definition: Data Mining

Data mining is the systematic application of computer-aided methods to discover patterns, trends or correlations in (often large and connected) data sets. Algorithms based on statistical methods are frequently used for this purpose.



Data mining

... is a useful toolkit if we care more about the *what* rather than the *why*:

- Data mining is often good at
 - ... making *predictions under stability*.
 - ... finding patterns in the data. These may foster research questions that we can examine further in other data sources.
- Data mining is less good at
 - ... helping to come up with a good theory or to improve an existing one.
 - ... producing valid inference. It has a tendency to find **false positives**.¹

[1] A false positive is a result that incorrectly indicates the presence of a condition under examination.



Data mining – predictions under stability

Example: War in Ukraine and Oil prices

- The global market price of crude oil was ~\$76 per barrel in January 2022.² The war drove prices to \$110 at around March 2022 and increased volatility substantially → an *unstable* DGP.
- Data mining methods tend to make bad forecasts given such DGP change and also fail to provide reliable measures of forecast uncertainty.
- Models based on economic and statistical theory better adapt to new (market) conditions.

[2] see <https://www.gep.com/blog/mind/russia-ukraine-wars-effects-oil-and-gas-industry#:~:text=Oil%20prices%20were%20rising%20globally,barrel%20on%204%20March%202022>.



Data mining – looking for patterns

Example: Viagra

To find a pattern can make you rich

- Pfizer scientists originally developed Viagra to mitigate high blood pressure and angina.
- Famous side effects have been found using data mining: instead of coming to the data with a theory, the scientists noticed an interesting pattern in the data.

Data mining is bad as a final step

- Taking a pattern found as given would be problematic
- Pfizer did clinical studies to *see if the pattern replicated*: they verified an hypothesis derived from their research question using experimental data.



So what makes a research question a *good* one?

Simplicity

- Do not bundle a bunch of research questions into one.
- Consider potential answers — research question and theory probably do not align very well if we cannot place potential answers within the framework of our theory.

Feasibility / scalability

- A good research question can be answered using the right data.
- Data acquisition is often subject to constraints — think time, money, data protection laws.

Suitable research design

- The research design must be suitable to investigate the derived hypothesis.
- Thoughtful consideration must be given as to which statistical method is appropriate. We will deal with this aspect in particular in this course.